

A. Problem, user, and purpose

1. **Who is our *primary* user?**
 - Asset management companies, commercial building owners wanting to make a change, retrofitting their properties for compliance or sustainability
 2. **Who are the *secondary* stakeholders we care about but don't design the UI for first?**
 - Mayor's office, their process is simplified and sustainability goals are pushed and met?
 3. **What is the single sentence problem statement?**
 - "Today, commercial building owners struggle with multiple steps they have to get done on retrofitting and compliance checking before reaching the mayor's office for approval.
 4. **What real world event triggers a user to come to us?**
 - Say that Ansley Mall is trying to do the retrofit job described, but they want to know where to get started in application process, what part of their building can they retrofit, how much difference is it gonna make it on their electricity efficiency etc, before they get to the mayors office
 5. **What decisions are they trying to make that are currently slow/expensive/confusing?**
 - They have an idea, but don't know how their retrofit will align with prerequisites for applications, and what benefits they will receive in terms of sustainability goals
 6. **What happens to them if they *don't* solve this (cost, fines, missed incentives, risk)?**
 - CBEEO fines, lost tax benefits, environmental/atl city ordinance non compliance, lost rebates, higher flood/heat risk, reputational/ESG downside.
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B. Policies and constraints

7. **Which ordinances and plans must our users comply with?**
 - CBEEO (benchmarking + ASHRAE Level II audits).
 - Post-Development Stormwater Ordinance (1-inch GI requirement).
 - Chapter 75 sustainable building policies
 - https://www.investatlanta.com/assets/c-pace_program_guidelines_-_jan_2025_k357L65.pdf
 - <https://www.atlantaga.gov/home/showdocument?id=59680&t=638269295768431125>
 - ENERGY STAR Score and Energy Use Index (EUI), Reporting in US EPA

8. **For each ordinance, what is the key “number” or rule we encode?**
 - CBEEO: thresholds (25k ft²), audit frequency, 15% savings for exemptions, fines.
 - Stormwater: 1.0” rainfall volume that must be managed on-site with GI.[[atlantawatershed](#)]
 - ENERGY STAR Score: 75 or higher for certification
 9. **Which of these rules do we actually enforce or visualize in the MVP, and which are just narrative?**
 - Chapter 75: Design exemptions and prerequisite information in sections 75-23 to 75-26.
 - C-PACE Program: Any of the eligibility, financial, and documentation information in 5-10.
 - Any of the rules from the permitting services guide will mostly be in 8-10 and 50-56, and follow up steps in 65-68.
 - Certified ENERGY STAR Scores and EUI
 10. **What project types are *in scope* for us?**
 - Retrofits / redevelopments that touch roofs and sites (not ground-up new construction, not tiny interior renovations).
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C. Data and inputs

11. **What datasets do we need at minimum to run the MVP?**
 - CBEEO building benchmarking excel export (ABID, address, floor area, commercial, housing).
 - Building footprints / roof area from google maps, google earth
 - GI & solar performance/cost factors.
12. **Which of those datasets do we already have, and which do we stub/mock?**

We have CBEEO building benchmarking excel export (ABID, address, floor area, commercial, housing).

Building footprints / roof area from google maps, google earth

A working repo for figma layout of building with intervention layers, rooftop solar, lot canopy solar, green roof, stormwater trees, beltline lattice.
13. **At runtime, what inputs does the user provide?**
 - e.g., building selection, budget band, preferred priority (ROI vs impact vs water).
*** stubbed to be buttons that always go to the same result, which will output the mall’s solar panels, parking lot shades, etc
14. **What assumptions are we making in the background (unit costs, performance factors) that we will document explicitly?**

{

"site_id": "ansley_mall_hero_001",
"site_name": "Ansley Mall",
"address": "1544 Piedmont Ave NE, Atlanta, GA 30324",
"city": "Atlanta",
"state": "GA",
"zip_code": "30324",
"neighborhood": "Piedmont Heights",
"site_type": "Open-air shopping mall / community retail center",
"owner": "Selig",
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"estimated_fields": {
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 "parking_lot_area_sf": 335200,
 "canopy_eligible_area_sf": 167600,
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"baseline_annual_electricity_kwh": 2975000,
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"baseline_annual_stormwater_target_gal_1in": 434900,
"existing_stormwater_capture_gal": 0,
"cbeco_applicability": true,
"energy_star_certification_threshold": 75
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"policy_flags": {
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  "ashrae_level_ii_audit_relevant": true,
  "stormwater_1in_requirement_relevant": true,
  "chapter_75_relevant": true,
  "c_pace_relevant": true,
  "energy_star_reporting_relevant": true
},

"hero_defaults": {
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  "priority_mode": "Balanced",
  "available_toggles": [
    "rooftop_solar",
    "parking_canopy_solar",
    "green_roof",
    "stormwater_trees",
    "bioswale_planters"
  ]
}
}

```

Large businesses

Any codes and regulations that are connected to the solar panels that are in the mall demo

D. Outputs and value

15. What *exactly* does the app show on screen?

- Interactive front end model that can toggle on and off the different retrofitted pieces, changing the outputs like prices, credits, etc for the suggested retrofit

- Views for both applicants/ asset companies/ commercial buildings and for the city of atl mayors sustainability
 - Ledger that has all the discussed pieces for moving the company forward with application process mentioned above
 - 16. **What *exactly* do we give the user as a file or “report”?**
 - Name, length, sections of the Retrofit Action Dossier.
 - 17. **What are the 3–5 key numbers they walk away with for a building?**
 - kWh/year saved, capex range, simple payback, gallons/year managed, % of 1” storm addressed, cost and time to build it
 - 18. **What decisions can they make *immediately* after using our tool?**
 - “Choose Package B for Building X, talk to the engineer and the city with this dossier.”
 - 19. **How do city staff benefit from those same outputs (even if they’re not the ones pressing the buttons)?**

They get another login screen where they can see what proactive measures they can take with existing building to reach sustainability goals, current applications, etc
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E. Scope and priorities

- 20. **Which user journey do we support *end-to-end* in the hackathon?**
 - Clearly describe: “Select building → see baseline → select toggles and see packages → export dossier.”
 - 21. **Which journeys are explicitly out of scope (for now)?**
 - e.g., permitting submission, contractor bidding platform, architect/designed platform
 - 22. **How many real buildings will we support in the MVP?**
 - One hero case(the mall demo)
 - 23. **What parts of the UI are live vs static mock?**
 - UI live would be the figma of the live demo mall, where user can toggle what to add etc,
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F. Architecture and implementation

- 24. **What is our minimal technical stack?**
 - Front-end technology, backend/API layer, data store.
- 25. **Where do we store configuration (GI costs, performance curves, policy thresholds)?**
 - Store backend for each user

26. **How do we generate the dossier: server-side PDF, HTML page, or slide template?**
- Server-side PDF
27. **What is our plan for faking or seeding data so the demo never breaks?**
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G. Validation and impact

28. **How will we demonstrate that what we recommend is *plausible* (not exact, but not nonsense)?**
- References to case studies, GI guidelines, PV tools, City of Atlanta's BenchmarkATL and Smart Surfaces Coalition initiatives, statistic of ENERGY STAR Certified buildings to all buildings in metro-Atlanta area
29. **What's the simplest quantitative story of impact we can tell for the hero building?**
- "This package saves ~\$X/year and manages Y,000 gallons of water, with Z-year payback."
 - "The proposed plan also allows business owners to identify the financial and traction incentives and potential."
 - "This district/building is a good candidate for the city/private investors to commit resources for a successful project."
30. **What feedback or "asks" do we want from judges / sponsors after they see the demo?**
- Data access, pilot buildings, policy input, funding.
 - Expansion to other cities with UHI and environmental justice concerns